Sergei Shorin (28 years old)

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Senior Go backend developer with 7+ years in software development. I design and run backend services with large data volumes (PostgreSQL, ClickHouse, Kafka) and ship risky changes in reversible stages. I use AI agents as a controlled tool, verifying their output with tests and real data. Passionate about mathematics and computer science, with a background in low-level C and assembly.

Work Experience

Senior Backend Developer (Go) — Ozon

Stack: Go, PostgreSQL, ClickHouse, Kafka, Elasticsearch, gRPC, Kubernetes, GitLab CI, Claude Code
August 2025 — Present (Senior since September 2026)

- Building a mass-hiring CRM: the candidate journey from application to onboarding (warehouse, couriers, support).
- **PostgreSQL and ClickHouse:** traced recurring database degradations using production metrics: retuned autovacuum, dropped 24 unused indexes (90 GB), moved heavy queries from SQL to Go with shadow comparison in production. Migrated 700M events to ClickHouse with zero downtime and no API changes (at-least-once, deduplication, reads merged from two stores). Result: service p99 783 → 364 ms, DB CPU down by 84%, database size 535 → 210 GB.
- **Email communication history service from scratch:** Kafka consumer, enrichment worker, sharded PostgreSQL, gRPC API; idempotent handling of duplicate and out-of-order events.
- **Legacy refactoring with AI agents:** replaced two incompatible slot-booking implementations (6 slot types) with a single model that enforces invariants at the database level. Proved correctness with 191 e2e scenarios and a staged rollout via runtime config (shadow async → sync → read switch), reconciled against real data. Result: legacy code removed, a smaller unified codebase, and fixed races and data inconsistencies the old architecture could not solve.
- **Generic async task engine:** shared PostgreSQL-backed queue, retries, batch and parallel processing; removed duplicated worker code, used by the team in 5+ processes.
- **Candidate workstation check system from scratch:** desktop app (Go + Wails, Windows/macOS), internet speed test microservice (WebSocket, load limits), CI/CD release process: tests, code signing and notarization, two approvals, rollback without a rebuild.
- **AI-assisted development in the team:** built a set of 10+ skills for AI agents (working on large tasks, local debugging, incident investigation, log search, migrations, tests, review) and train the team to work with agents.
- **Product development and support:** new vacancy launches and integrations, bulk Elasticsearch reindexing mechanism, incident investigations (lost update, cron timeouts), on-call duty, migration of services to PostgreSQL 17.

Backend / Full-stack Developer — Funexpected LLC

Stack: Python, MongoDB, AWS, Docker, GitHub Actions, C++, C, Objective-C, GDScript, Godot Engine
March 2019 — August 2025 (6.5 years) iOS, site

- Educational app for children aged 3-8 (App Store / Google Play): developed the backend (hosted on AWS), infrastructure and client.
- **Accounts and saves:** account and save system, including merging saves made during simultaneous play; family accounts with multiple children and school accounts with classes.
- **Statistics:** service that collects and processes in-game statistics; reports for parents and a separate website for teachers with class statistics.
- **On-demand content distribution:** the app downloads only the courses and languages it needs.
- **Infrastructure:** made build export 76x faster by optimizing file system operations, automated app uploads to the App Store, built an A/B testing system.
- **ML:** handwriting recognition (CRNN + libonnx) — 98% accuracy; added the missing layers to the lightweight libonnx library, so the app size grew by only 3 MB.
- **Gamedev:** 15 mini-games and a unified educational course, procedural level generation (10 games); animations and shaders — the project received *The Webby Award* for visual design.

Education

Bachelor of Applied Mathematics and Data Science — HSE University, Moscow
September 2021 — June 2025. Thesis: using AI agents to develop large software systems (case study: Generia, Go microservices)

Ozon Route 256 (Go, Middle level) —

June — July 2025

École 42 (21 coding school by Sberbank) —
February 2019 — August 2023

Yandex Backend Development School (C++) —
June — July 2024 — userver, microservices, testing, databases, DevOps, monitoring

Tech Stack

- **Languages:** Go, C++, C, Python
- **Backend:** Kafka, gRPC, microservices
- **Databases:** PostgreSQL, ClickHouse, Elasticsearch, MongoDB
- **DevOps:** AWS, Kubernetes, Docker, CI/CD (GitLab CI, GitHub Actions)
- **Tools:** Git, Linux, Claude Code (AI agents)
- **Engine:** Godot Engine (GDScript)

Projects (selection)

Backend / Highload

- **Generia** — HSE graduation project: ”Instagram for virtual worlds”, Go + Microservices + LLM + Stable Diffusion. code
- **HTTP WebServer** — C++ HTTP/1.1 server with CGI, handles 3k rps. code

Data Science / DL

- **number_CRNN** — PyTorch. Recognition of handwritten numbers and signs. CRNN architecture. code
- **dl-image-classification** — PyTorch, WandB; classification of 200 classes. code
- **en-ru-translation** — Transformer + PyTorch, WandB, Hydra + FastAPI translation demo service. code

Low-Level

- **ssl-md5** — Implementation of MD5, SHA-2, DES, Base64; OOP in C. code
- **Snake-RISC-V** — Game on RISC-V ASM, memory-mapped I/O, graphics. code
- **libonnx-fork** — ONNX engine in C; Added recurrent layer, dynamic quantization. code

GameDev / Graphics

- **Game Engine** — C++ engine with CPU renderer and animations. code

Additional Mini-Projects

- **con-run** — P2P cryptocurrency node (Go, Gossip/PEX). code
- **cv-latex-builder** — CI/CD for LaTeX resume. code
- + More pet projects on GitHub

Languages

Russian — Native | English — B2 (Intermediate-High)